

EXTRA PROBLEM ABOUT DUMMIES AND INTERACTION TERMS

ECON301—ECONOMETRICS I

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As I said in a previous announcement, a study plan for the exam involves understanding *all topics from lectures 4, 5, 6, and 7 (this includes the blackboard slides, the videos, and the practical examples that I explain during the video lectures) in addition to all the optional homework problems I talked about during the lectures and problem sets 3, 4 and 5.*

However, for the topics of dummy variables, interaction terms, and quadratic terms you did not have the chance to work on a problem of your own. This extra problem is meant to help you with that. I will not upload solutions to this problem set. You must work on it on your own.

Problem:

What are important determinants on student's academic achievement? There are many determinants of this, but to answer part of this question, a cross-section data set was constructed using a population of US high school students, in which information about the student's gender, race and size of high school graduating class was collected. The researchers are trying to estimate the following BLP:

$$E^*[sat|hsize, hsize^2, female, black, female \times black] = \beta_0 + \beta_1 hsize + \beta_2 hsize^2 + \beta_3 female + \beta_4 black + \beta_5 female \times black \quad (1)$$

where the variables are:

- *sat* is the combined SAT score;
- *hsize* is the size of the student's high school graduating class, in hundreds;
- *female* is a gender dummy variable;
- *black* is a race dummy variable equal to one for blacks, and zero otherwise.

The estimation results are in Table 1. The standard errors are in parenthesis, below the estimated coefficient. The estimate for the variance covariance matrix is provided in table 2.

Questions:

1. Is there strong evidence that $hsize^2$ should be included in the model? From this equation, what is the optimal high school size?
2. Holding *hsize* fixed, what is the estimated difference in SAT score between nonblack females and nonblack males? how statistically significant is this estimated difference?
3. What is the estimated difference in SAT score between nonblack males and black males? Test the null hypothesis that there is no difference between their scores, against the alternative that there is a difference.
4. What is the estimated difference in SAT score between black females and nonblack females? What would you need to do to test whether the difference is statistically significant?
5. Consider student *A*, which graduated from a high school class of 300 ($hsize = 3$), is female and not black. Student *B* graduated from a high school class of 400 ($hsize = 4$) and is also female and not black. Let $\Delta sat = sat_B - sat_A$ denote the difference in SAT scores where sat_B and sat_A are the SAT scores of students B and A respectively. Delineate a procedure that will allow you to construct a 95% confidence interval around Δsat . Use your proposed procedure to estimate the confidence interval.

Table 1: Estimation results from the model (1)

<i>Dependent variable:</i>	
	sat
hsize	19.297*** (3.814)
hsizesq	−2.195*** (0.524)
female	−45.091*** (4.226)
black	−169.813*** (15.423)
female×black	62.306*** (19.305)
Constant	1,028.097*** (6.355)
Observations	4,137
R ²	0.086
Adjusted R ²	0.085
Note:	*p<0.1; **p<0.05; ***p<0.01

Table 2: Estimated Variance-Covariance Matrix

$$\hat{\Omega} = \begin{bmatrix} 40.386516 & -20.199194 & 2.335095 & -9.031893 & -11.536296 & 8.210525 \\ -20.199194 & 14.545411 & -1.900289 & 0.078404 & 1.261833 & 0.068763 \\ 2.335095 & -1.900289 & 0.274387 & -0.000038 & -0.077895 & 0.031187 \\ -9.031893 & 0.078404 & -0.000038 & 17.863001 & 8.820194 & -17.816754 \\ -11.536296 & 1.261833 & -0.077895 & 8.820194 & 237.863505 & -237.367652 \\ 8.210525 & 0.068763 & 0.031187 & -17.816754 & -237.367652 & 372.702161 \end{bmatrix}$$